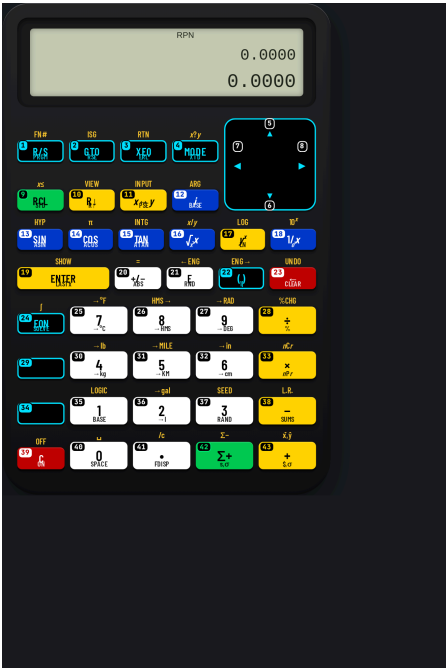
U-45 CALCULATORS

User Manual

Calculator #22 RPN Scientific Programmable (2-line, black)

Scientific – RPN



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#22 — RPN Scientific Programmable (2-line, black)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example 

. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

RPN with a 4-level stack (ENTER separates numbers, operators act at once).

Chapters

1. Getting started
2. Features
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5. Worked examples
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7. Keyboard shortcuts
8. Common mistakes
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1 · Getting started

Open U-45 CALCULATORS, choose **#22** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#20, #21, #25, #26, #27, #30, #31, #32, #35, #36, #37, #40, #41
op	Operators & execute	#10, #11, #17, #19, #28, #33, #38, #43
fn	Functions	#12, #13, #14, #15, #16, #18
mem	Memory, variables & registers	#9, #42
clr	Clear, delete & power	#23, #39
mod	Shift, modes, menus & navigation	#1, #2, #3, #4, #5, #6, #7, #8, #22, #24, #29, #34

2 · Features

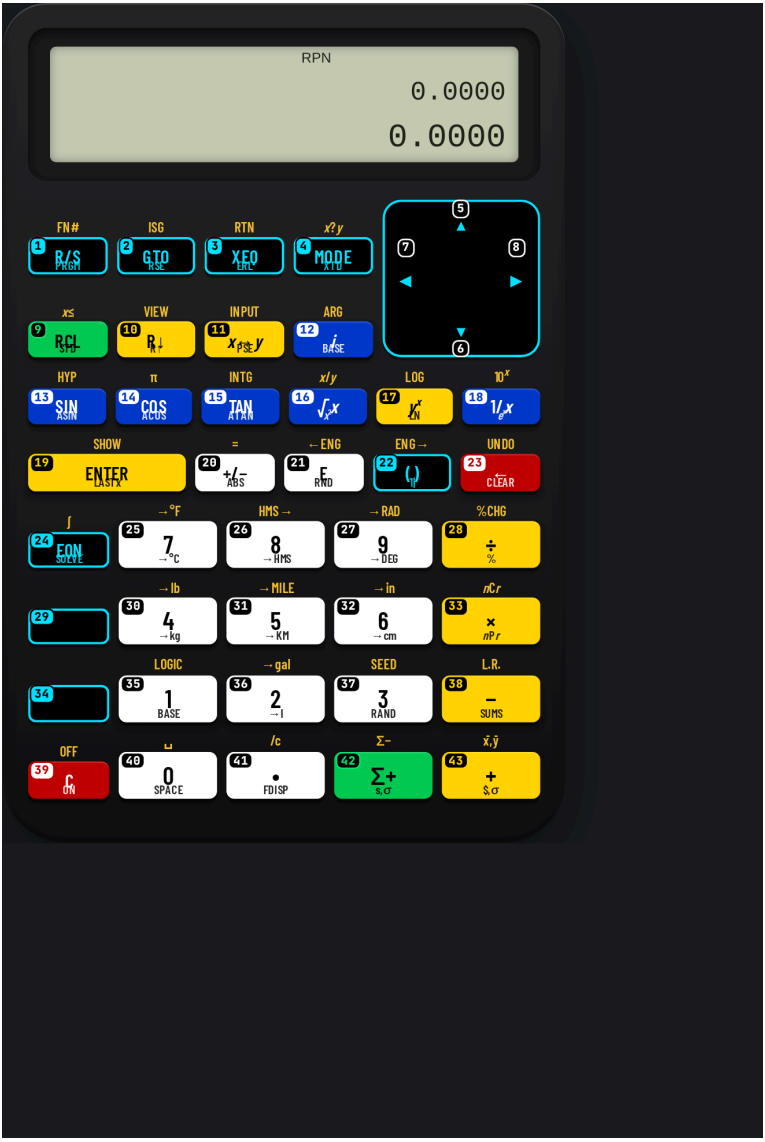
45 features: 26 live on the keys, 9 app/UI features, 10 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **RPN stack with 2-line (Y over X) display**
2. LIVE **Yellow / blue shifts**
3. LIVE **Numbered menus: MODE, DISPLAY, CLEAR, BASE, LOGIC, L.R., SUMS, s,σ, INTG, PROB**
4. LIVE **Complex i entry**
5. LIVE **→HR / →RAD conversions**
6. LIVE **Linear regression ŷ, x̂, r**
7. LIVE **SHOW full precision**
8. LIVE **Square root √**
9. LIVE **Any-index root $\sqrt[x]{y}$**
10. LIVE **Reciprocal x^{-1} (1/x)**
11. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
12. LIVE **Common logarithm $\log_{10}$**
13. LIVE **Antilog 10^x**
14. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
15. LIVE **Combinations nCr**
16. LIVE **Percent change Δ% between two values**
17. LIVE **π constant to full internal precision**
18. LIVE **Scientific-notation entry ×10^x / EXP / EEX**
19. LIVE **Sign change (−) / +/− / CHS**

20. LIVE **Engineering notation ENG (exponent multiple of 3)**
21. LIVE **Recall memory / variables (RCL, MR, recall)**
22. LIVE **Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2**
23. LIVE **Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular**
24. LIVE **Bitwise logic AND / OR / XOR / XNOR / NOT**
25. LIVE **Backspace / DEL editing of the current entry**
26. LIVE **Exchange x ⇄ y / x ⇄ M**
27. APP/UI **Every key numbered 1…N in reading order — matches the key map in the user manual**
28. APP/UI **High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle**
29. APP/UI **Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)**
30. APP/UI **Show / hide key numbers with one button (remembered between sessions)**
31. APP/UI **Copy the current display text to the clipboard**
32. APP/UI **Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
33. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
34. APP/UI **Reference photo viewer and tool-tip name on every key**
35. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
36. PLANNED **Equation list**
37. PLANNED **Programs**
38. PLANNED **SOLVE**

39. PLANNED ∫ integration
40. PLANNED Unit conversions (°F, lb)
41. PLANNED Stack history view
42. PLANNED Undo
43. PLANNED Equation library
44. PLANNED Speech read-out
45. PLANNED Colour-blind key theme

3 · Numbered key map



Calculator #22 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	R/S	FN#	·	·	mod
2	GTO	ISG	·	·	mod
3	XEQ	RTN	·	·	mod
4	MODE	x?y	·	·	mod
5	▲	·	·	·	mod
6	▼	·	·	·	mod
7	◀	·	·	·	mod
8	▶	·	·	·	mod
9	RCL	x≤	·	·	mem
10	R↓	VIEW	·	·	op
11	x ↔ y	INPUT	·	·	op
12	i	ARG	·	·	fn
13	SIN	HYP	·	·	fn
14	COS	π	·	·	fn
15	TAN	INTG	·	·	fn
16	√x	x/y	·	·	fn
17	yx	LOG	·	·	op
18	1/x	10x	·	·	fn
19	ENTER	SHOW	·	·	op
20	+/−	=	·	·	num
21	E	← ENG	·	·	num
22	( )	ENG →	·	·	mod
23	←	UNDO	·	·	clr

#	Face	Shift	Alpha	3rd	Group
24	EQN	∫	·	·	mod
25	7	→°F	·	·	num
26	8	HMS→	·	·	num
27	9	→RAD	·	·	num
28	÷	%CHG	·	·	op
29	shift	·	·	·	mod
30	4	→lb	·	·	num
31	5	→MILE	·	·	num
32	6	→in	·	·	num
33	×	nCr	·	·	op
34	gshift	·	·	·	mod
35	1	LOGIC	·	·	num
36	2	→gal	·	·	num
37	3	SEED	·	·	num
38	−	L.R.	·	·	op
39	C	OFF	·	·	clr
40	0	␣	·	·	num
41	•	/c	·	·	num
42	Σ+	Σ−	·	·	mem
43	+	x̄,ȳ	·	·	op

4 · Key-by-key reference

Every key on calculator #22, in number order, with every function it carries in this app.

#1 R/S `Shift, modes, menus & navigation`

Primary: Shows the message “Programs n/a” (feature not provided in this app)

shifted (2nd / f / SHIFT layer) (press #29 first): FN# — Shows the message “Programs n/a” (feature not provided in this app)

Printed under the key: PRGM

#2 GTO `Shift, modes, menus & navigation`

Primary: Shows the message “Programs n/a” (feature not provided in this app)

Printed under the key: RSE

#3 XEQ `Shift, modes, menus & navigation`

Primary: Shows the message “Programs n/a” (feature not provided in this app)

Printed under the key: ERL

#4 MODE `Shift, modes, menus & navigation`

Primary: MODE menu

Printed under the key: XTD

#5 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#6 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#7 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#8 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#9 **RCL** `Memory, variables & registers`

Primary: RCL — recall a memory / variable

Printed under the key: STD

#10 **R↓** `Operators & execute`

Primary: R↓ — roll the stack down

Printed under the key: R↑

#11 **x ↔ y** `Operators & execute`

Primary: x≷y — exchange X and Y

Printed under the key: PSE

#12 i Functions

Primary: Imaginary unit i / complex entry

shifted (2nd / f / SHIFT layer) (press #29 first): ARG — Shows the message "ARG: use →POL" (feature not provided in this app)

Printed under the key: BASE

#13 SIN Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #29 first): HYP — Shows the message "HYP: press SHIFT HYP then SIN" (feature not provided in this app)

Printed under the key: ASIN

Keyboard: S

#14 COS Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #29 first): π — π (3.14159265…) as an exact constant

Printed under the key: ACOS

Keyboard: C

#15 TAN Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #29 first): INTG — Integration

Printed under the key: ATAN

Keyboard: t

#16 √x Functions

Primary: Square root √

shifted (2nd / f / SHIFT layer) (press #29 first): x/y — x-th root of y

Printed under the key: x^2

Keyboard: r

#17 yx Operators & execute

Primary: y^x — raise Y to the power X (RPN: base first, then exponent)

shifted (2nd / f / SHIFT layer) (press #29 first): LOG — Common logarithm $\log_{10}$

Printed under the key: LN

Keyboard: ^

#18 1/x Functions

Primary: Reciprocal x^{-1} (1/x)

shifted (2nd / f / SHIFT layer) (press #29 first): 10x — 10^x — antilog

Printed under the key: ex

Keyboard: i

#19 ENTER `Operators & execute`

Primary: ENTER — separate two numbers (push X to Y)

shifted (2nd / f / SHIFT layer) (press #29 first): SHOW — SHOW — full-precision view of X

Printed under the key: LASTx

Keyboard: `Enter`

#20 +/− `Digits & number entry`

Primary: Change sign (+/−) of the number being typed or the current result

Printed under the key: ABS

#21 E `Digits & number entry`

Primary: EEX — enter a power-of-ten exponent for the number in X

shifted (2nd / f / SHIFT layer) (press #29 first): ←ENG — ENG — engineering notation (exponent multiple of 3)

Printed under the key: RND

Keyboard: `e`

#22 ( ) `Shift, modes, menus & navigation`

Primary: Printed label only (no action in this app)

shifted (2nd / f / SHIFT layer) (press #29 first): ENG→ — ENG — engineering notation (exponent multiple of 3)

Printed under the key: 1|

#23 ← Clear, delete & power

Primary: Backspace — delete the last digit typed

Printed under the key: CLEAR

Keyboard: Backspace Delete

#24 EQN Shift, modes, menus & navigation

Primary: Shows the message “Equations n/a” (feature not provided in this app)

shifted (2nd / f / SHIFT layer) (press #29 first): ∫ — Shows the message “∫ needs an equation” (feature not provided in this app)

Printed under the key: SOLVE

#25 7 Digits & number entry

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #29 first): →°F — Shows the message “→°F: use CONVERT” (feature not provided in this app)

Printed under the key: →°C

Keyboard: 7

#26 8 Digits & number entry

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #29 first): HMS→ — →HR — h.ms to hours

Printed under the key: →HMS

Keyboard: 8

#27 9 Digits & number entry

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #29 first): →RAD — →RAD — degrees to radians

Printed under the key: →DEG

Keyboard: 9

#28 ÷ Operators & execute

Primary: Division ÷

shifted (2nd / f / SHIFT layer) (press #29 first): %CHG — Δ% — percent change

Printed under the key: %

Keyboard: /

#29 shift Shift, modes, menus & navigation

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#30 4 Digits & number entry

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #29 first): →lb — Shows the message "→lb n/a" (feature not provided in this app)

Printed under the key: →kg

Keyboard: 4

#31 5 Digits & number entry

Primary: Digit 5

Printed under the key: → KM

Keyboard: 5

#32 6 Digits & number entry

Primary: Digit 6

Printed under the key: → cm

Keyboard: 6

#33 × Operators & execute

Primary: Multiplication ×

shifted (2nd / f / SHIFT layer) (press #29 first): nCr — Combinations / permutations

Printed under the key: nPr

Keyboard: * x

#34 gshift Shift, modes, menus & navigation

Primary: g — blue prefix: next key does its lower function

#35 1 `Digits & number entry`

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #29 first): LOGIC — Logic functions

Printed under the key: BASE

Keyboard: 1

#36 2 `Digits & number entry`

Primary: Digit 2

Printed under the key: →I

Keyboard: 2

#37 3 `Digits & number entry`

Primary: Digit 3

Printed under the key: RAND

Keyboard: 3

#38 – `Operators & execute`

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #29 first): L.R. — Statistics/regression menu

Printed under the key: SUMS

Keyboard: -

#39 C Clear, delete & power

Primary: CLx — clear X

shifted (2nd / f / SHIFT layer) (press #29 first): OFF — OFF — power off (state is lost; press ON to restart)

Printed under the key: ON

#40 0 Digits & number entry

Primary: Digit 0

Printed under the key: SPACE

Keyboard: 0

#41 • Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Printed under the key: FDISP

Keyboard: . ,

#42 Σ+ Memory, variables & registers

Primary: Σ+ — enter a statistics data point

shifted (2nd / f / SHIFT layer) (press #29 first): Σ− — Σ− — remove a statistics data point

Printed under the key: s,σ

#43 + `Operators & execute`

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #29 first): x̄,ȳ — x̄ — mean

Printed under the key: $,σ

Keyboard: +

5 · Worked examples

16 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#35 1 → #36 2 → #19 ENTER → #37 3 → #19 ENTER → #30 4 → #33 × → #43 +

Display shows 24.0000 **— expected 24. ✔ verified on the app.**

RPN needs no precedence rules: the product is formed first because you key it first, then + adds it to the number waiting on the stack.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#35 1 → #36 2 → #19 ENTER → #26 8 → #43 + → #30 4 → #28 ÷

Display shows 5.0000 **— expected 5. ✔ verified on the app.**

In RPN the brackets disappear — add first, then divide.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#37 3 → #19 ENTER → #30 4 → #43 + → #27 9 → #19 ENTER → #36 2 → #38 − → #33 ×

Display shows 49.0000 **— expected 49. ✔ verified on the app.**

Each bracket becomes its own stack result; the final × combines the two.

Example 5.4 — Square roots

Calculate √144. Press Reset first.

#35 1 → #30 4 → #30 4 → #16 √x

Display shows 12.0000 **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.5 — Powers

Calculate 2^10. Press Reset first.

#36 2 → #19 ENTER → #35 1 → #40 0 → #17 y^x

Display shows 1024.0000 **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.6 — Reciprocals

Calculate 1 ÷ 4 (x^{-1}). Press Reset first.

`#30` 4 → `#18` 1/*x*

Display shows 0.2500 **— expected 0.25. ✔ verified on the app.**

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.7 — Trigonometry (degrees)

Calculate sin 30° (Deg mode). Press Reset first.

`#37` 3 → `#40` 0 → `#13` SIN

Display shows 0.5000 **— expected 0.5. ✔ verified on the app.**

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.8 — Common logarithm

Calculate log 1000. Press Reset first.

`#35` 1 → `#40` 0 → `#40` 0 → `#40` 0 → `#29` → `#17` LOG

Display shows 3.0000 **— expected 3. ✔ verified on the app.**

log is base-10: it counts the zeros of a power of ten.

Example 5.9 — Antilog 10^x

Calculate 10^2. Press Reset first.

#36 2 → #29 → #18 10^x

Display shows 100.0000 **— expected 100. ✔ verified on the app.**

10^x is the inverse of log.

Example 5.10 — Negative numbers

Calculate −5 + 12. Press Reset first.

#31 5 → #20 +/− → #19 ENTER → #35 1 → #36 2 → #43 +

Display shows 7.0000 **— expected 7. ✔ verified on the app.**

Use the sign key for a negative number — not the subtraction key.

Example 5.11 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

#32 6 → #21 E → #36 2 → #19 ENTER → #37 3 → #33 ×

Display shows 1800.0000 **— expected 1800. ✔ verified on the app.**

The exponent key enters $\times 10^n$ without typing × 10 ^.

Example 5.12 — Continue from the answer

Calculate 12 × 3 = , then Ans ÷ 5. Press Reset first.

#35 1 → #36 2 → #19 ENTER → #37 3 → #33 × → #31 5 → #28 ÷

Display shows 7.2000 **— expected 7.2. ✔ verified on the app.**

The result simply stays in X, ready for the next operation.

Example 5.13 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#30 4 → #19 ENTER → #26 8 → #43 + → #35 1 → #31 5 → #43 + → #37 3 → #28 ÷

Display shows 9.0000 **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.14 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#31 5 → #40 0 → #19 ENTER → #30 4 → #19 ENTER → #32 6 → #33 × → #38 − → #25 7 → #43 +

Display shows 33.0000 **— expected 33. ✔ verified on the app.**

Key the product first (it waits on the stack), subtract it, then add the last value.

Example 5.15 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#37 3 → #32 6 → #40 0 → #19 ENTER → #30 4 → #28 ÷ → #27 9 → #28 ÷

Display shows 10.0000 **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.16 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#37 3 → #19 ENTER → #31 5 → #43 + → #30 4 → #33 × → #36 2 → #38 − → #35 1 → #40 0 → #28 ÷

Display shows 3.0000 **— expected 3. ✔ verified on the app.**

Work from the innermost bracket outwards — each step leaves its result in X.

6 · Tricks & hacks

Hack 1: No equals key

RPN works number, number, operator: 3 ENTER 4 + = 7. Results stay on the stack for the next step.

Hack 2: Chain without ENTER

After an operation, just type the next number and operator: 3 ENTER 4 + 5 × gives (3+4)×5 = 35.

Hack 3: Swap to fix order

If you entered numbers in the wrong order for − ÷ or y^x, press x≷y before the operator.

Hack 4: LASTx to reuse

LASTx brings back the X value of the previous operation — useful for a × a.

Hack 5: Stack as memory

Up to four values are kept; R↓ rolls through them.

Hack 6: Constant via stack top

On 4-level stacks the T register copies down, so a constant multiplier can be reused by pressing × repeatedly.

Hack 7: Hidden functions on this model

23 extra functions sit on shifted layers. Press the layer key once, then the key: for example #29 then #1 → FN#; #29 then #12 → ARG; #29 then #13 → HYP; #29 then #14 → π.

Hack 8: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #43.

Hack 9: Type faster on a computer

24 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 10: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 11: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 12: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
s	#13	SIN	Sine (uses the current angle unit)
c	#14	COS	Cosine (current angle unit)
t	#15	TAN	Tangent (current angle unit)
r	#16	√x	Square root √
^	#17	yx	y^x — raise Y to the power X (RPN: base first, then exponent)
i	#18	1/x	Reciprocal x^{-1} (1/x)
Enter	#19	ENTER	ENTER — separate two numbers (push X to Y)
e	#21	E	EEX — enter a power-of-ten exponent for the number in X
Backspace Delete	#23	←	Backspace — delete the last digit typed
7	#25	7	Digit 7
8	#26	8	Digit 8
9	#27	9	Digit 9
/	#28	÷	Division ÷
4	#30	4	Digit 4
5	#31	5	Digit 5
6	#32	6	Digit 6
* x	#33	×	Multiplication ×
1	#35	1	Digit 1
2	#36	2	Digit 2
3	#37	3	Digit 3
-	#38	–	Subtraction –

0	#40	0	Digit 0
. ,	#41	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
+	#43	+	Addition +

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Pressing ENTER after an operation

ENTER after + duplicates the result. Usually you want to type the next number directly.

Mistake 2: Wrong order for – and ÷

Y – X and Y ÷ X: enter the first operand first. 10 ENTER 4 – = 6.

Mistake 3: CHS vs –

Make a number negative with CHS/+/–, not with –.

Mistake 4: Prefix left active

If f / g / SHIFT is shown, the next key is shifted. Clear it with the prefix-clear function.

Mistake 5: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 6: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (101)

Work each one on calculator #22, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 2.5 + 4 × 1.5 = _____
7. 1000 + 25 × 4 = _____
8. 19 + 21 × 3 = _____
9. 64 + 8 × 8 = _____
10. 0.5 + 0.25 × 8 = _____
11. (45 + 15) ÷ 6 = _____
12. (9 + 7) ÷ 2 = _____
13. (100 + 60) ÷ 8 = _____
14. (33 + 17) ÷ 5 = _____
15. (7.5 + 2.5) ÷ 4 = _____
16. (120 + 80) ÷ 16 = _____

17. (1 + 2) ÷ 4 = _____

18. (250 + 350) ÷ 12 = _____

19. (5 + 6) × (10 − 3) = _____

20. (12 + 3) × (8 − 5) = _____

21. (1.5 + 2.5) × (7 − 4) = _____

22. (20 + 30) × (15 − 5) = _____

23. (9 + 1) × (100 − 1) = _____

24. (11 + 13) × (17 − 9) = _____

25. √625 = _____

26. √2 = _____

27. √50 = _____

28. √1024 = _____

29. √3 = _____

30. √0.25 = _____

31. √9801 = _____

32. 3^5 = _____

33. 1.05^3 = _____

34. 5^4 = _____

35. 7^3 = _____

36. 1.5^2 = _____

37. 10^6 = _____

38. 2^0.5 = _____

39. 1 ÷ 8 (x^{-1}) = _____

40. 1 ÷ 0.25 (x^{-1}) = _____

41. 1 ÷ 16 (x^{-1}) = _____

42. 1 ÷ 5 (x^{-1}) = _____

43. 1 ÷ 40 (x^{-1}) = _____

44. 1 ÷ 0.2 (x^{-1}) = _____

45. cos 60° (Deg mode) = _____

46. tan 45° (Deg mode) = _____

47. sin 90° (Deg mode) = _____

48. cos 0° (Deg mode) = _____

49. log 100 = _____

50. log 10000 = _____

51. 10^3 = _____

52. −20 + 7 = _____

53. −3 + 3 = _____

54. −15.5 + 4 = _____

55. −100 + 250 = _____

56. −8 + 1 = _____

57. 2×10^3 × 4 = _____

58. 1.5×10^4 × 2 = _____

59. 3×10^5 × 3 = _____

60. 4.5×10^2 × 2 = _____

61. 100 × 4 = , then Ans ÷ 2 = _____

62. 15 × 6 = , then Ans ÷ 9 = _____

63. 7.5 × 8 = , then Ans ÷ 3 = _____

64. 250 × 12 = , then Ans ÷ 100 = _____

65. (12 + 15 + 18) ÷ 3 = _____

66. (70 + 85 + 91) ÷ 3 = _____

67. (2.5 + 3.5 + 6) ÷ 3 = _____

68. (100 + 200 + 600) ÷ 3 = _____

69. 100 − 3 × 12 + 9 = _____

70. 18 − 2 × 5 + 40 = _____

71. 7.5 − 1.5 × 3 + 2 = _____

72. 64 − 8 × 4 + 10 = _____

73. 300 − 25 × 4 + 55 = _____

74. 9 − 3 × 3 + 1 = _____

75. 45 − 5 × 5 + 15 = _____

76. 1000 ÷ 8 ÷ 5 = _____

77. 96 ÷ 2 ÷ 3 = _____

78. 7200 ÷ 60 ÷ 60 = _____

79. 500 ÷ 4 ÷ 25 = _____

80. 81 ÷ 3 ÷ 3 = _____

81. 1.5 ÷ 0.5 ÷ 2 = _____

82. ((10 + 2) × 3 − 6) ÷ 6 = _____

83. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

84. ((20 + 30) × 2 − 25) ÷ 3 = _____

85. ((7 + 8) × 9 − 15) ÷ 12 = _____

86. 12 + 3 × 4 = _____

87. (12 + 8) ÷ 4 = _____

88. (3 + 4) × (9 − 2) = _____

89. √144 = _____

90. 2^10 = _____

91. 1 ÷ 4 (x^{-1}) = _____

92. sin 30° (Deg mode) = _____

93. log 1000 = _____

94. 10^2 = _____

95. −5 + 12 = _____

96. 6×10^2 × 3 = _____

97. 12 × 3 = , then Ans ÷ 5 = _____

98. (4 + 8 + 15) ÷ 3 = _____

99. 50 − 4 × 6 + 7 = _____

100. 360 ÷ 4 ÷ 9 = _____

101. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (55)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “R/S” (Shows the message “Programs n/a”)? _____
2. Which key is “GTO” (Shows the message “Programs n/a”)? _____
3. Which key is “XEQ” (Shows the message “Programs n/a”)? _____
4. Which key is “MODE” (MODE menu)? _____
5. Which key is “▲” (Cursor ▲)? _____
6. Which key is “▼” (Cursor ▼)? _____
7. Which key is “◀” (Cursor ◀)? _____
8. Which key is “▶” (Cursor ▶)? _____
9. Which key is “RCL” (RCL)? _____
10. Which key is “R↓” (R↓)? _____
11. Which key is “x↔y” (x≷y)? _____
12. Which key is “i” (Imaginary unit i / complex entry)? _____
13. Which key is “SIN” (Sine)? _____

14. Which key is “COS” (Cosine)? _____

15. Which key is “TAN” (Tangent)? _____

16. Which key is “√x” (Square root √)? _____

17. Which key is “yx” (y^x)? _____

18. Which key is “1/x” (Reciprocal x^{-1})? _____

19. Which key is “ENTER” (ENTER)? _____

20. Which key is “+/–” (Change sign)? _____

21. Which key is “E” (EEX)? _____

22. Which key is “ ← ” (Backspace)? _____

23. Which key is “EQN” (Shows the message “Equations n/a”)? _____

24. Which key is “÷” (Division ÷)? _____

25. Which key is “shift” (SHIFT / 2nd)? _____

26. Which key is “×” (Multiplication ×)? _____

27. Which key is “gshift” (g)? _____

28. Which key is “–” (Subtraction –)? _____

29. Which key is “C” (CLx)? _____

30. Which key is “•” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

31. Which key is “Σ+” (Σ+)? _____

32. Which key is “+” (Addition +)? _____

33. How do you reach “FN#”? _____

34. How do you reach “ARG”? _____

35. How do you reach “HYP”? _____

36. How do you reach “π”? _____

37. How do you reach “INTG”? _____

38. How do you reach “x/y”? _____

39. How do you reach “LOG”? _____

40. How do you reach “10x”? _____

41. How do you reach “SHOW”? _____

42. How do you reach “ ← ENG”? _____

43. How do you reach “ENG → ”? _____

44. How do you reach “∫”? _____

45. How do you reach “ → °F”? _____

46. How do you reach “HMS → ”? _____

47. How do you reach “ → RAD”? _____

48. How do you reach “%CHG”? _____

49. How do you reach “ → lb”? _____

50. How do you reach “nCr”? _____

51. How do you reach “LOGIC”? _____

52. How do you reach “L.R.”? _____

53. How do you reach “OFF”? _____

54. How do you reach “∑–”? _____

55. How do you reach “x̄,ȳ”? _____

Appendix A — Drill answers

Set A

1. 55 (display `55.0000`)

`#36` 2 → `#31` 5 → `#19` ENTER → `#31` 5 → `#19` ENTER → `#32` 6 → `#33` × → `#43` +

2. 79 (display `79.0000`)

`#25` 7 → `#19` ENTER → `#26` 8 → `#19` ENTER → `#27` 9 → `#33` × → `#43` +

3. 282 (display `282.0000`)

`#35` 1 → `#31` 5 → `#40` 0 → `#19` ENTER → `#35` 1 → `#36` 2 → `#19` ENTER → `#35` 1 → `#35` 1 → `#33` × → `#43` +

4. 213 (display `213.0000`)

`#37` 3 → `#19` ENTER → `#35` 1 → `#30` 4 → `#19` ENTER → `#35` 1 → `#31` 5 → `#33` × → `#43` +

5. 90 (display `90.0000`)

`#30` 4 → `#26` 8 → `#19` ENTER → `#32` 6 → `#19` ENTER → `#25` 7 → `#33` × → `#43` +

6. 8.5 (display `8.5000`)

`#36` 2 → `#41` • → `#31` 5 → `#19` ENTER → `#30` 4 → `#19` ENTER → `#35` 1 → `#41` • → `#31` 5 → `#33` × → `#43` +

7. 1100 (display `1100.0000`)

#35 1 → #40 0 → #40 0 → #40 0 → #19 ENTER → #36 2 → #31 5 → #19 ENTER → #30 4 → #33 × → #43 +

8. 82 (display `82.0000`)

#35 1 → #27 9 → #19 ENTER → #36 2 → #35 1 → #19 ENTER → #37 3 → #33 × → #43 +

9. 128 (display `128.0000`)

#32 6 → #30 4 → #19 ENTER → #26 8 → #19 ENTER → #26 8 → #33 × → #43 +

10. 2.5 (display `2.5000`)

#40 0 → #41 • → #31 5 → #19 ENTER → #40 0 → #41 • → #36 2 → #31 5 → #19 ENTER → #26 8 → #33 × → #43 +

11. 10 (display `10.0000`)

#30 4 → #31 5 → #19 ENTER → #35 1 → #31 5 → #43 + → #32 6 → #28 ÷

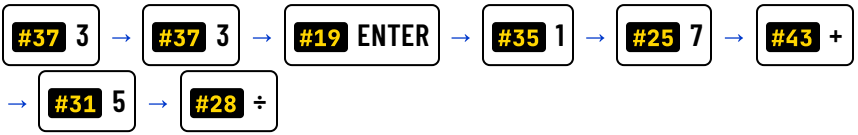
12. 8 (display `8.0000`)

#27 9 → #19 ENTER → #25 7 → #43 + → #36 2 → #28 ÷

13. 20 (display `20.0000`)

#35 1 → #40 0 → #40 0 → #19 ENTER → #32 6 → #40 0 → #43 + → #26 8 → #28 ÷

14. 10 (display `10.0000`)



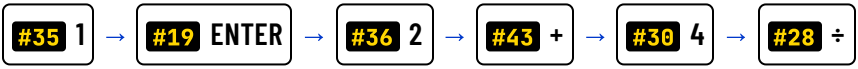
15. 2.5 (display `2.5000`)



16. 12.5 (display `12.5000`)



17. 0.75 (display `0.7500`)



18. 50 (display `50.0000`)



19. 77 (display `77.0000`)



20. 45 (display `45.0000`)



21. 12 (display `12.0000`)

`#35` 1 → `#41` • → `#31` 5 → `#19` ENTER → `#36` 2 → `#41` • → `#31` 5 → `#43` + → `#25` 7 → `#19` ENTER → `#30` 4 → `#38` − → `#33` ×

22. 500 (display `500.0000`)

`#36` 2 → `#40` 0 → `#19` ENTER → `#37` 3 → `#40` 0 → `#43` + → `#35` 1 → `#31` 5 → `#19` ENTER → `#31` 5 → `#38` − → `#33` ×

23. 990 (display `990.0000`)

`#27` 9 → `#19` ENTER → `#35` 1 → `#43` + → `#35` 1 → `#40` 0 → `#40` 0 → `#19` ENTER → `#35` 1 → `#38` − → `#33` ×

24. 192 (display `192.0000`)

`#35` 1 → `#35` 1 → `#19` ENTER → `#35` 1 → `#37` 3 → `#43` + → `#35` 1 → `#25` 7 → `#19` ENTER → `#27` 9 → `#38` − → `#33` ×

25. 25 (display `25.0000`)

`#32` 6 → `#36` 2 → `#31` 5 → `#16` $\sqrt{x}$

26. 1.41421356 (display `1.4142`)

`#36` 2 → `#16` $\sqrt{x}$

27. 7.07106781 (display `7.0711`)

`#31` 5 → `#40` 0 → `#16` $\sqrt{x}$

28. 32 (display `32.0000`)

#35 1 → #40 0 → #36 2 → #30 4 → #16 √x

29. 1.73205081 (display `1.7321`)

#37 3 → #16 √x

30. 0.5 (display `0.5000`)

#40 0 → #41 • → #36 2 → #31 5 → #16 √x

31. 99 (display `99.0000`)

#27 9 → #26 8 → #40 0 → #35 1 → #16 √x

32. 243 (display `243.0000`)

#37 3 → #19 ENTER → #31 5 → #17 y^x

33. 1.157625 (display `1.1576`)

#35 1 → #41 • → #40 0 → #31 5 → #19 ENTER → #37 3 → #17 y^x

34. 625 (display `625.0000`)

#31 5 → #19 ENTER → #30 4 → #17 y^x

35. 343 (display `343.0000`)

#25 7 → #19 ENTER → #37 3 → #17 y^x

36. 2.25 (display `2.2500`)

#35 1 → #41 • → #31 5 → #19 ENTER → #36 2 → #17 y^x

37. 1000000.000 (display `1000000.0000`)

`#35` 1 → `#40` 0 → `#19` ENTER → `#32` 6 → `#17` y^x

38. 1.41421356 (display `1.4142`)

`#36` 2 → `#19` ENTER → `#40` 0 → `#41` • → `#31` 5 → `#17` y^x

39. 0.125 (display `0.1250`)

`#26` 8 → `#18` 1/*x*

40. 4 (display `4.0000`)

`#40` 0 → `#41` • → `#36` 2 → `#31` 5 → `#18` 1/*x*

41. 0.0625 (display `0.0625`)

`#35` 1 → `#32` 6 → `#18` 1/*x*

42. 0.2 (display `0.2000`)

`#31` 5 → `#18` 1/*x*

43. 0.025 (display `0.0250`)

`#30` 4 → `#40` 0 → `#18` 1/*x*

44. 5 (display `5.0000`)

`#40` 0 → `#41` • → `#36` 2 → `#18` 1/*x*

45. 0.5 (display `0.5000`)

`#32` 6 → `#40` 0 → `#14` COS

46. 1 (display `1.0000`)

`#30` 4 → `#31` 5 → `#15` TAN

47. 1 (display `1.0000`)

`#27` 9 → `#40` 0 → `#13` SIN

48. 1 (display `1.0000`)

`#40` 0 → `#14` COS

49. 2 (display `2.0000`)

`#35` 1 → `#40` 0 → `#40` 0 → `#29` → `#17` LOG

50. 4 (display `4.0000`)

`#35` 1 → `#40` 0 → `#40` 0 → `#40` 0 → `#40` 0 → `#29` → `#17` LOG

51. 1000 (display `1000.0000`)

`#37` 3 → `#29` → `#18` 10^x

52. -13 (display `-13.0000`)

`#36` 2 → `#40` 0 → `#20` +/− → `#19` ENTER → `#25` 7 → `#43` +

53. 0 (display `0.0000`)

`#37` 3 → `#20` +/− → `#19` ENTER → `#37` 3 → `#43` +

54. -11.5 (display `-11.5000`)

`#35` 1 → `#31` 5 → `#41` • → `#31` 5 → `#20` +/− → `#19` ENTER → `#30` 4 → `#43` +

55. 150 (display `150.0000`)

`#35` 1 → `#40` 0 → `#40` 0 → `#20` +/− → `#19` ENTER →
`#36` 2 → `#31` 5 → `#40` 0 → `#43` +

56. -7 (display `-7.0000`)

`#26` 8 → `#20` +/− → `#19` ENTER → `#35` 1 → `#43` +

57. 8000 (display `8000.0000`)

`#36` 2 → `#21` E → `#37` 3 → `#19` ENTER → `#30` 4 → `#33` ×

58. 30000 (display `30000.0000`)

`#35` 1 → `#41` • → `#31` 5 → `#21` E → `#30` 4 → `#19` ENTER
→ `#36` 2 → `#33` ×

59. 900000 (display `900000.0000`)

`#37` 3 → `#21` E → `#31` 5 → `#19` ENTER → `#37` 3 → `#33` ×

60. 900 (display `900.0000`)

`#30` 4 → `#41` • → `#31` 5 → `#21` E → `#36` 2 → `#19` ENTER
→ `#36` 2 → `#33` ×

61. 200 (display `200.0000`)

`#35` 1 → `#40` 0 → `#40` 0 → `#19` ENTER → `#30` 4 → `#33` ×
→ `#36` 2 → `#28` ÷

62. 10 (display `10.0000`)

`#35` 1 → `#31` 5 → `#19` ENTER → `#32` 6 → `#33` × → `#27` 9
→ `#28` ÷

63. 20 (display `20.0000`)

#25 7 → #41 • → #31 5 → #19 ENTER → #26 8 → #33 × → #37 3 → #28 ÷

64. 30 (display `30.0000`)

#36 2 → #31 5 → #40 0 → #19 ENTER → #35 1 → #36 2 → #33 × → #35 1 → #40 0 → #40 0 → #28 ÷

65. 15 (display `15.0000`)

#35 1 → #36 2 → #19 ENTER → #35 1 → #31 5 → #43 + → #35 1 → #26 8 → #43 + → #37 3 → #28 ÷

66. 82 (display `82.0000`)

#25 7 → #40 0 → #19 ENTER → #26 8 → #31 5 → #43 + → #27 9 → #35 1 → #43 + → #37 3 → #28 ÷

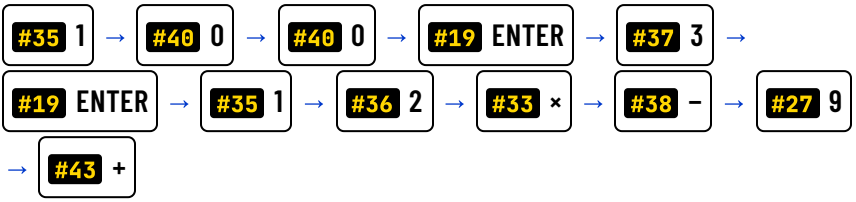
67. 4 (display `4.0000`)

#36 2 → #41 • → #31 5 → #19 ENTER → #37 3 → #41 • → #31 5 → #43 + → #32 6 → #43 + → #37 3 → #28 ÷

68. 300 (display `300.0000`)

#35 1 → #40 0 → #40 0 → #19 ENTER → #36 2 → #40 0 → #40 0 → #43 + → #32 6 → #40 0 → #40 0 → #43 + → #37 3 → #28 ÷

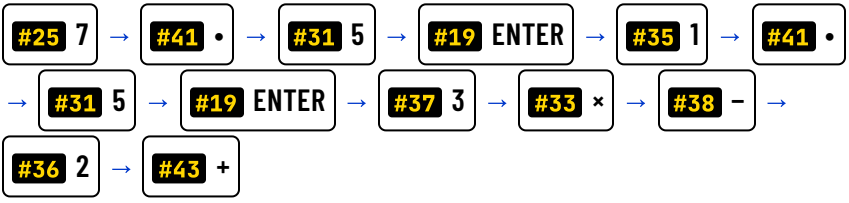
69. 73 (display `73.0000`)



70. 48 (display `48.0000`)



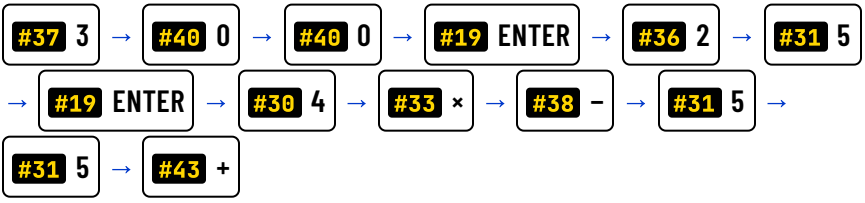
71. 5 (display `5.0000`)



72. 42 (display `42.0000`)



73. 255 (display `255.0000`)



74. 1 (display `1.0000`)



75. 35 (display `35.0000`)

`#30` 4 → `#31` 5 → `#19` ENTER → `#31` 5 → `#19` ENTER →
`#31` 5 → `#33` × → `#38` − → `#35` 1 → `#31` 5 → `#43` +

76. 25 (display `25.0000`)

`#35` 1 → `#40` 0 → `#40` 0 → `#40` 0 → `#19` ENTER → `#26` 8
→ `#28` ÷ → `#31` 5 → `#28` ÷

77. 16 (display `16.0000`)

`#27` 9 → `#32` 6 → `#19` ENTER → `#36` 2 → `#28` ÷ → `#37` 3
→ `#28` ÷

78. 2 (display `2.0000`)

`#25` 7 → `#36` 2 → `#40` 0 → `#40` 0 → `#19` ENTER → `#32` 6
→ `#40` 0 → `#28` ÷ → `#32` 6 → `#40` 0 → `#28` ÷

79. 5 (display `5.0000`)

`#31` 5 → `#40` 0 → `#40` 0 → `#19` ENTER → `#30` 4 → `#28` ÷
→ `#36` 2 → `#31` 5 → `#28` ÷

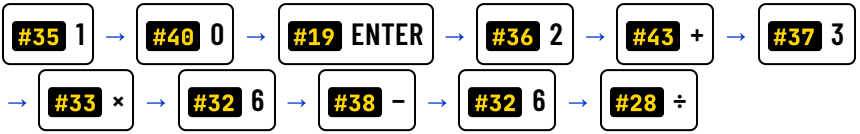
80. 9 (display `9.0000`)

`#26` 8 → `#35` 1 → `#19` ENTER → `#37` 3 → `#28` ÷ → `#37` 3
→ `#28` ÷

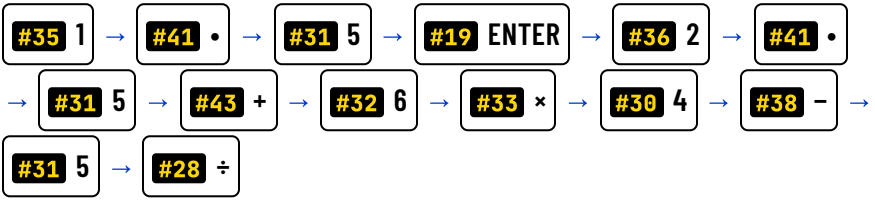
81. 1.5 (display `1.5000`)

`#35` 1 → `#41` • → `#31` 5 → `#19` ENTER → `#40` 0 → `#41` •
→ `#31` 5 → `#28` ÷ → `#36` 2 → `#28` ÷

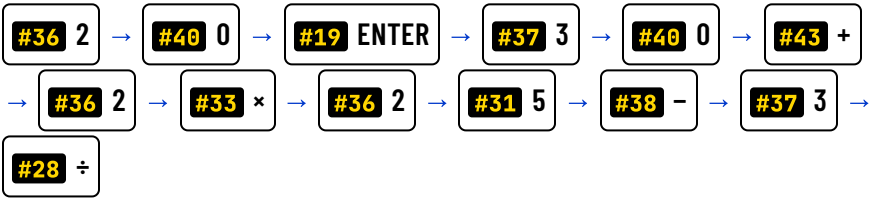
82. 5 (display `5.0000`)



83. 4 (display `4.0000`)



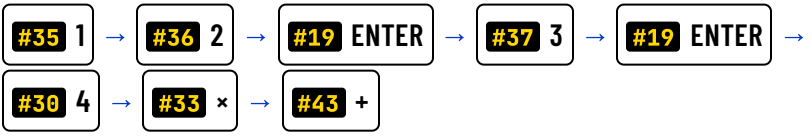
84. 25 (display `25.0000`)



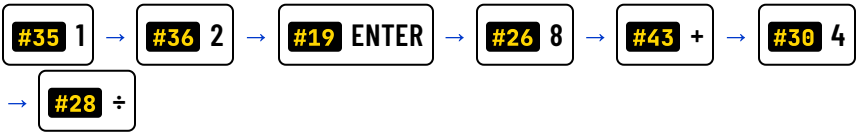
85. 10 (display `10.0000`)



86. 24 (display `24.0000`)



87. 5 (display `5.0000`)



88. 49 (display `49.0000`)



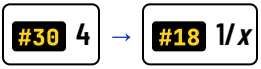
89. 12 (display `12.0000`)



90. 1024 (display `1024.0000`)



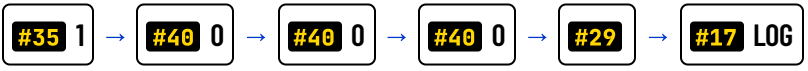
91. 0.25 (display `0.2500`)



92. 0.5 (display `0.5000`)



93. 3 (display `3.0000`)



94. 100 (display `100.0000`)



95. 7 (display `7.0000`)



96. 1800 (display `1800.0000`)



97. 7.2 (display 7.2000)

#35 1 → #36 2 → #19 ENTER → #37 3 → #33 × → #31 5 → #28 ÷

98. 9 (display 9.0000)

#30 4 → #19 ENTER → #26 8 → #43 + → #35 1 → #31 5 → #43 + → #37 3 → #28 ÷

99. 33 (display 33.0000)

#31 5 → #40 0 → #19 ENTER → #30 4 → #19 ENTER → #32 6 → #33 × → #38 − → #25 7 → #43 +

100. 10 (display 10.0000)

#37 3 → #32 6 → #40 0 → #19 ENTER → #30 4 → #28 ÷ → #27 9 → #28 ÷

101. 3 (display 3.0000)

#37 3 → #19 ENTER → #31 5 → #43 + → #30 4 → #33 × → #36 2 → #38 − → #35 1 → #40 0 → #28 ÷

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18

19. #19
20. #20
21. #21
22. #23
23. #24
24. #28
25. #29
26. #33
27. #34
28. #38
29. #39
30. #41
31. #42
32. #43
33. #29 then #1 R/S
34. #29 then #12 i
35. #29 then #13 SIN
36. #29 then #14 COS
37. #29 then #15 TAN
38. #29 then #16 √x
39. #29 then #17 yx
40. #29 then #18 1/x
41. #29 then #19 ENTER
42. #29 then #21 E
43. #29 then #22 ( )
44. #29 then #24 EQN
45. #29 then #25 7
46. #29 then #26 8
47. #29 then #27 9
48. #29 then #28 ÷
49. #29 then #30 4
50. #29 then #33 ×
51. #29 then #35 1
52. #29 then #38 -
53. #29 then #39 C
54. #29 then #42 Σ+
55. #29 then #43 +

Appendix B — Function finder (A–Z)

Function	Label	Keys
→ HR — h.ms to hours	HMS →	#29 → #26
→ RAD — degrees to radians	→ RAD	#29 → #27
10^x — antilog	10x	#29 → #18
Addition +	+	#43
Backspace — delete the last digit typed	←	#23
Change sign (+/−) of the number being typed or the current result	+/−	#20
CLx — clear X	C	#39
Combinations / permutations	nCr	#29 → #33
Common logarithm $\log_{10}$	LOG	#29 → #17
Cosine (current angle unit)	COS	#14
Cursor ▲ — up / previous history line / previous item	▲	#5
Cursor ▶ — move right in the line	▶	#8
Cursor ▼ — down / next history line / next item	▼	#6
Cursor ◀ — move left in the line	◀	#7
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	•	#41
Digit 0	0	#40
Digit 1	1	#35
Digit 2	2	#36
Digit 3	3	#37
Digit 4	4	#30
Digit 5	5	#31

Digit 6	6	#32
Digit 7	7	#25
Digit 8	8	#26
Digit 9	9	#27
Division ÷	÷	#28
EEX — enter a power-of-ten exponent for the number in X	E	#21
ENG — engineering notation (exponent multiple of 3)	← ENG	#29 → #21
ENG — engineering notation (exponent multiple of 3)	ENG →	#29 → #22
ENTER — separate two numbers (push X to Y)	ENTER	#19
g — blue prefix: next key does its lower function	gshift	#34
Imaginary unit i / complex entry	i	#12
Integration	INTG	#29 → #15
Logic functions	LOGIC	#29 → #35
MODE menu	MODE	#4
Multiplication ×	×	#33
OFF — power off (state is lost; press ON to restart)	OFF	#29 → #39
Printed label only (no action in this app)	( )	#22
R↓ — roll the stack down	R↓	#10
RCL — recall a memory / variable	RCL	#9
Reciprocal x^{-1} (1/x)	1/x	#18
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	shift	#29
SHOW — full-precision view of X	SHOW	#29 → #19
Shows the message "→°F: use CONVERT" (feature not provided in this app)	→°F	#29 → #25

Shows the message "→lb n/a" (feature not provided in this app)	→lb	#29 → #30
Shows the message "∫ needs an equation" (feature not provided in this app)	∫	#29 → #24
Shows the message "ARG: use →POL" (feature not provided in this app)	ARG	#29 → #12
Shows the message "Equations n/a" (feature not provided in this app)	EQN	#24
Shows the message "HYP: press SHIFT HYP then SIN" (feature not provided in this app)	HYP	#29 → #13
Shows the message "Programs n/a" (feature not provided in this app)	R/S	#1
Shows the message "Programs n/a" (feature not provided in this app)	FN#	#29 → #1
Shows the message "Programs n/a" (feature not provided in this app)	GTO	#2
Shows the message "Programs n/a" (feature not provided in this app)	XEQ	#3
Sine (uses the current angle unit)	SIN	#13
Square root √	√x	#16
Statistics/regression menu	L.R.	#29 → #38
Subtraction –	–	#38
Tangent (current angle unit)	TAN	#15
$\bar{x}$ — mean	$\bar{x}$,$\bar{y}$	#29 → #43
x-th root of y	x/y	#29 → #16
x≷y — exchange X and Y	x ↔ y	#11
y^x — raise Y to the power X (RPN: base first, then exponent)	yx	#17
Δ% — percent change	%CHG	#29 → #28
π (3.14159265…) as an exact constant	π	#29 → #14
Σ+ — enter a statistics data point	Σ+	#42

Σ− — remove a statistics data point	**Σ−**	**#29 → #42**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

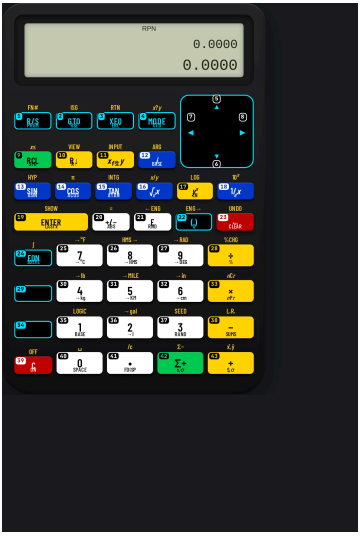
RPN with 2-line display (Y over X). Yellow (left) and blue (right) shifts; MODE, DISPLAY, CLEAR, BASE, LOGIC, L.R., SUMS, s,σ, INTG and PROB menus show numbered choices – press the digit. Unit conversions not wired are shown as labels only. Equations, programs, SOLVE and ∫ are not simulated.

Planned (not yet working)

- Equation list
- Programs
- SOLVE
- ∫ integration
- Unit conversions (°F, lb)
- Stack history view
- Undo
- Equation library
- Speech read-out
- Colour-blind key theme

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 R/S ⇧ FN#	#2 GTO ⇧ ISG	#3 XEQ ⇧ RTN
#4 MODE ⇧ x?y	#5 ▲	#6 ▼
#7 ◀	#8 ▶	#9 RCL ⇧ x≤
#10 R↓ ⇧ VIEW	#11 x ↔ y ⇧ INPUT	#12 i ⇧ ARG
#13 SIN ⇧ HYP	#14 COS ⇧ π	#15 TAN ⇧ INTG
#16 √x ⇧ x/y	#17 yx ⇧ LOG	#18 1/x ⇧ 10x
#19 ENTER ⇧ SHOW	#20 +/− ⇧ =	#21 E ⇧ ←ENG
#22 ( ) ⇧ ENG→	#23 ← ⇧ UNDO	#24 EQN ⇧ ∫
#25 7 ⇧ →°F	#26 8 ⇧ HMS→	#27 9 ⇧ →RAD
#28 ÷ ⇧ %CHG	#29 shift	#30 4 ⇧ →lb
#31 5 ⇧ →MILE	#32 6 ⇧ →in	#33 × ⇧ nCr
#34 gshift	#35 1 ⇧ LOGIC	#36 2 ⇧ →gal
#37 3 ⇧ SEED	#38 − ⇧ L.R.	#39 C ⇧ OFF
#40 0 ⇧ ␣	#41 • ⇧ /c	#42 Σ+ ⇧ Σ−
#43 + ⇧ x̄,ȳ		

U-45 CALCULATORS · Calculator #22 · cut out and keep